



SCHOOL OF FUTURE TECH

Case Study Report

On

Harbor Lane Legal Group

Network Foundations, Topology & IP
Addressing Design

Computer Networks

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Computer Networking Project Report

Network Foundations, Topology & IP Addressing Design

Case Study: Enterprise Network Design and Implementation

Project Title:

Design and Implementation of a Secure Enterprise Network Infrastructure

Organization:

Harbor Lane Legal Group

Subject:

Computer Networking

Tools Used:

Cisco Packet Tracer

Technologies Implemented:

- VLAN Segmentation
 - VLSM IP Addressing
 - Router-on-a-Stick Inter-VLAN Routing
 - DHCP Configuration
 - Voice VLAN
 - Trunking
 - WAN Leased Line Connectivity
 - Network Verification and Testing
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5	Network Requirements Analysis
6	Physical Network Topology Design
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Abstract

This project focuses on designing and implementing a complete enterprise network infrastructure for Harbor Lane Legal Group. The organization requires a reliable and scalable network connecting its Head Office and two branch offices: Riverside and Uptown.

The proposed solution divides the network into multiple logical segments using Virtual Local Area Networks (VLANs). Each department is assigned a dedicated VLAN to improve security, reduce broadcast traffic, and simplify network management.

A VLSM-based addressing scheme is implemented to efficiently allocate IP addresses according to department requirements. Router-on-a-Stick configuration is used to enable communication between different VLANs, while trunk links allow multiple VLANs to pass between network devices.

The network also includes WAN connectivity between offices using serial leased lines, DHCP-based IP allocation for VoIP devices, and complete verification through Cisco IOS commands and connectivity testing.

The complete design was simulated and tested using Cisco Packet Tracer.

1. Introduction

Modern organizations require secure, scalable, and reliable communication networks to support daily operations. Legal organizations especially handle sensitive information, requiring proper network segmentation, controlled access, and reliable connectivity between departments and offices.

Harbor Lane Legal Group requires a network infrastructure that supports multiple departments, employee devices, printers, VoIP communication, and branch office connectivity.

The objective of this project is to design a professional enterprise network architecture that provides:

- Secure departmental separation
- Efficient IP address utilization
- Reliable communication
- Easy troubleshooting
- Future scalability

The network consists of:

- One Head Office
- Riverside Branch Office
- Uptown Branch Office

The Head Office contains four departments:

1. Litigation Department
2. Corporate Law Department
3. IT & Records Department
4. Reception/Admin Department

Each department is separated using VLAN technology.

2. Problem Statement

Harbor Lane Legal Group is expanding its operations and requires a properly designed network before purchasing networking equipment.

The network design must provide:

- Separate networks for different departments
- Communication between departments when required
- Connectivity between Head Office and branch offices
- Proper IP addressing scheme
- Support for printers and VoIP devices
- Secure and manageable infrastructure

The network should minimize unnecessary broadcast traffic while maintaining efficient communication.

3. Project Objectives

The major objectives of this project are:

3.1 Network Segmentation

Each department is assigned a separate VLAN to isolate traffic and improve security.

3.2 Efficient IP Address Management

VLSM is used to allocate IP addresses according to actual host requirements.

3.3 Inter-VLAN Communication

Router-on-a-Stick configuration enables communication between different VLAN networks.

3.4 WAN Connectivity

The Head Office is connected with Riverside and Uptown branches through serial WAN links.

3.5 Voice Communication

A dedicated Voice VLAN is configured for IP phone communication.

3.6 Network Verification

The complete implementation is tested using:

- Ping testing
- Routing table verification
- DHCP verification
- Interface status checking

4. Network Requirements Analysis

The required network segments are:

Network Segment	Required Hosts	Purpose
Litigation Department	45	Attorney systems and workstations
Corporate Law Department	33	Corporate legal employees

IT & Records Department	9	IT resources and records management
Reception/Admin Department	4	Reception devices and administration
Voice Network	IP Phones	VoIP communication
Riverside WAN Link	2	Point-to-point connection
Uptown WAN Link	2	Point-to-point connection

5. Physical Network Topology Design

The implemented network follows an **Extended Star Topology**.

In this design, a central Core Switch connects multiple departmental access switches. The Head Office Router provides inter-VLAN routing and WAN connectivity.

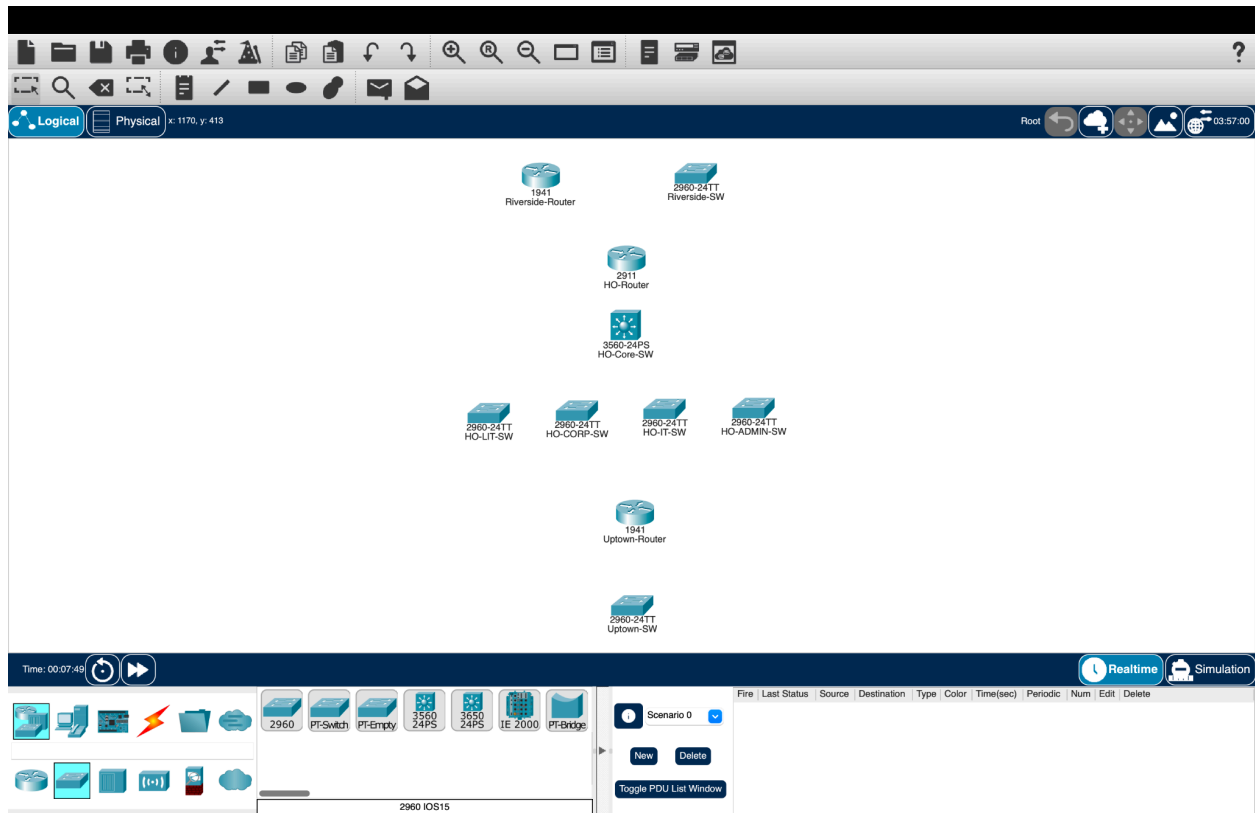
The Core Switch acts as the backbone device responsible for forwarding traffic between different network segments.

The access switches provide connectivity to:

- PCs
 - Laptops
 - Printers
 - IP Phones
-

5.1 Base Network Device Placement

The initial placement of routers, switches, and end devices was performed according to the required enterprise architecture.



Device: Packet Tracer Logical Workspace

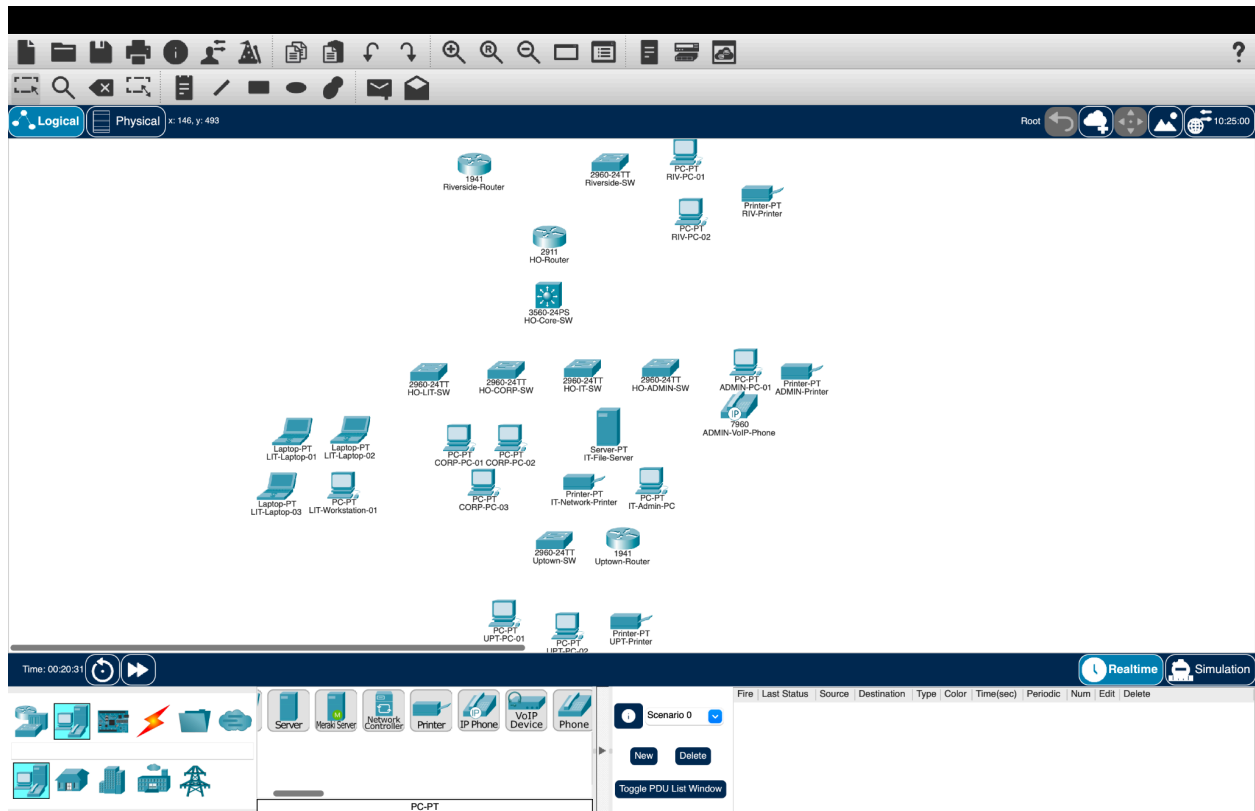
Purpose: Shows initial placement of all networking devices.

5.2 End Device Placement

All departmental end devices were placed according to organizational requirements.

Devices include:

- Employee laptops
- Desktop computers
- Printers
- IP Phone



Device: All Department End Devices

Purpose: Shows complete end-user device arrangement.

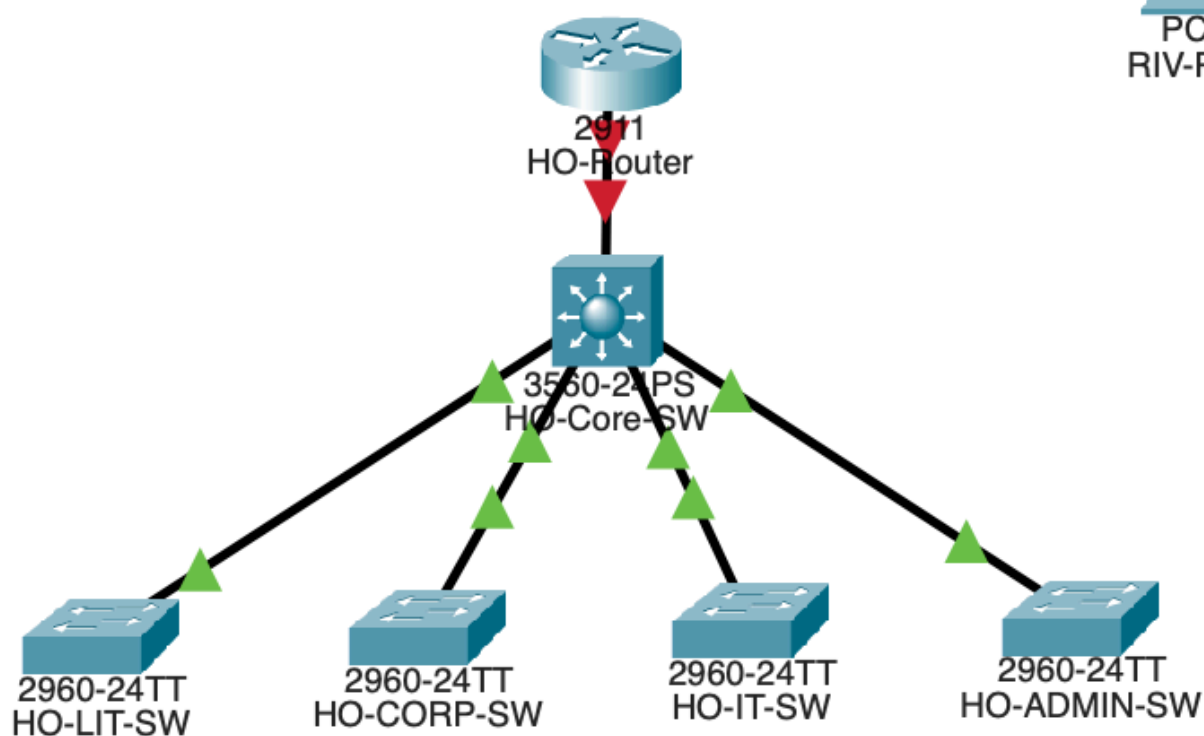
6. Head Office Backbone Design

The Head Office uses a centralized backbone architecture.

The Core Switch connects:

- Litigation Switch
- Corporate Law Switch
- IT Records Switch
- Reception/Admin Switch
- Head Office Router

The backbone links use trunk connections to carry multiple VLAN traffic.

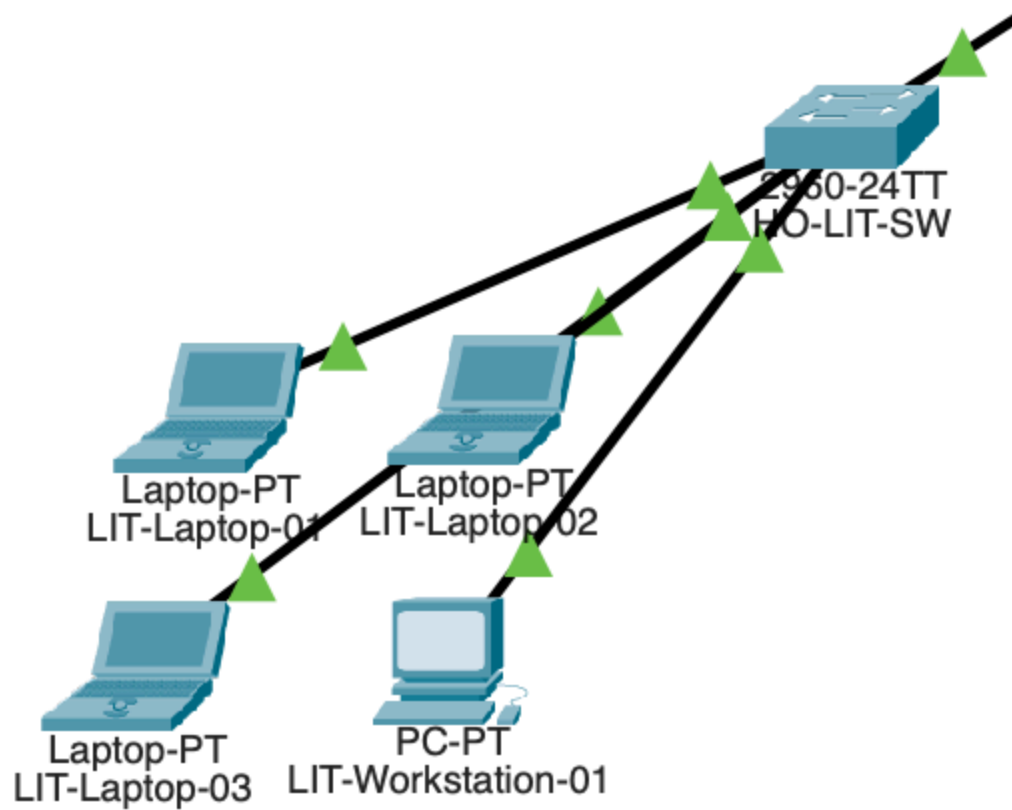


Device: HO Core Network

Purpose: Shows backbone connectivity between core devices.

6.1 Litigation Department Network

The Litigation Department contains employee devices connected through a dedicated access switch.

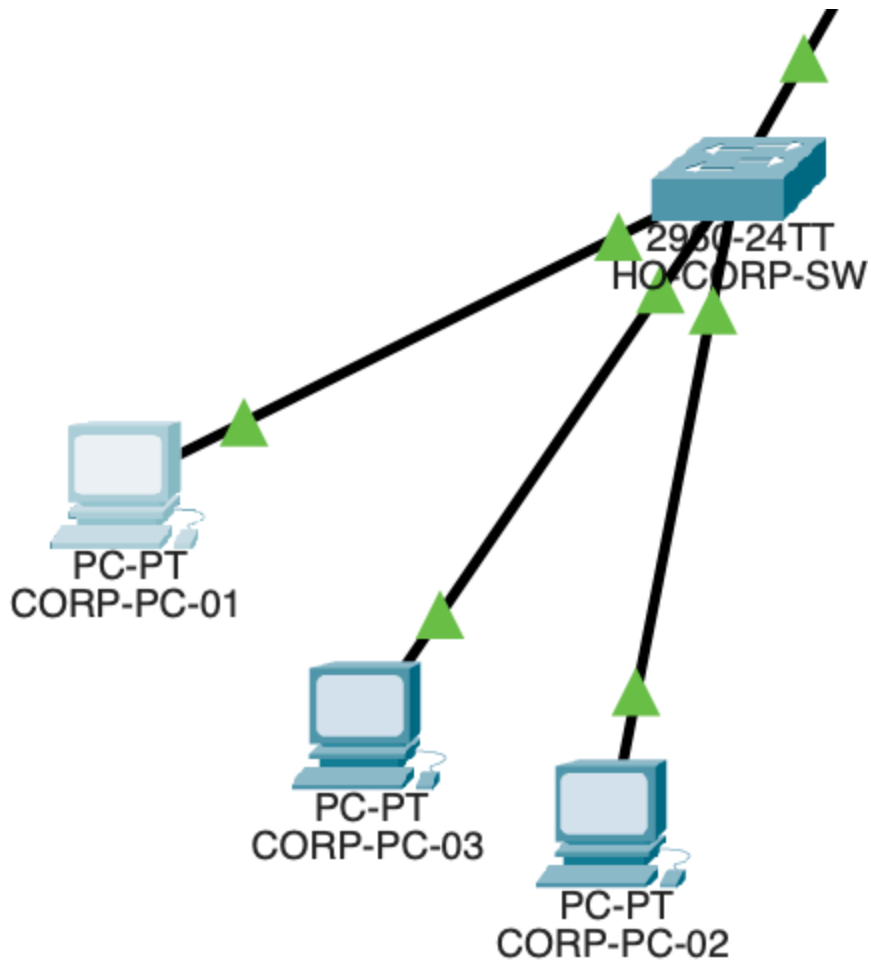


Device: HO-LIT-SW

Purpose: Shows Litigation department physical connections.

6.2 Corporate Law Department Network

The Corporate Law department has a separate network segment to maintain traffic isolation.

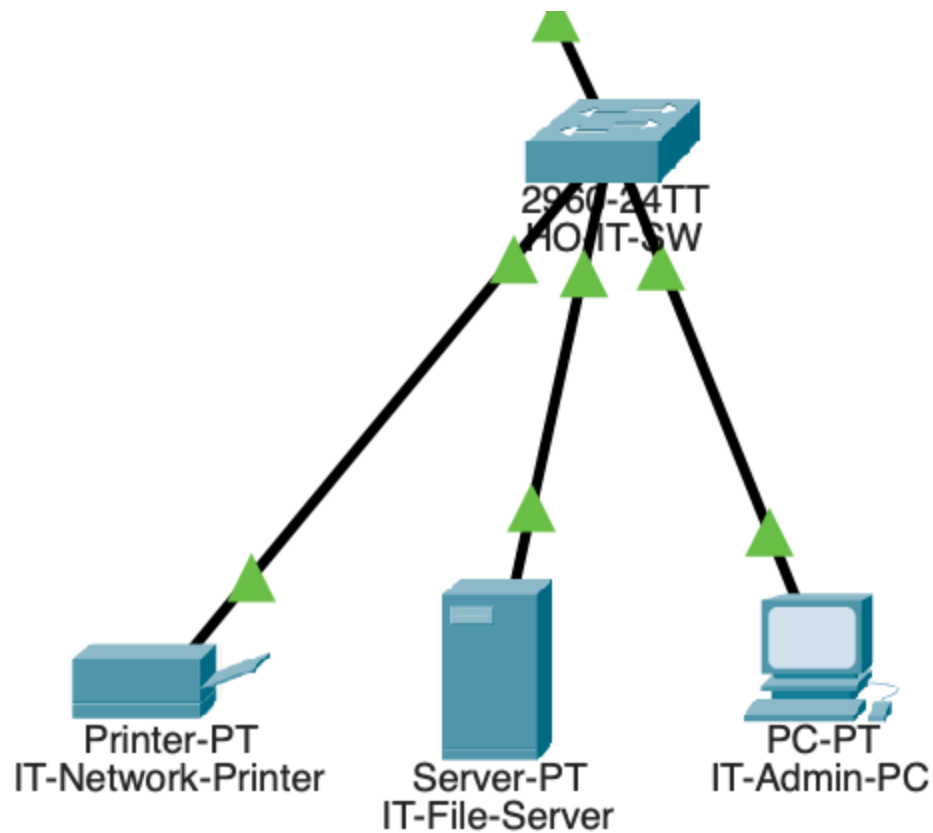


Device: HO-CORP-SW

Purpose: Shows Corporate Law department connections.

6.3 IT & Records Department Network

The IT and Records department contains systems responsible for technical resources and network management.

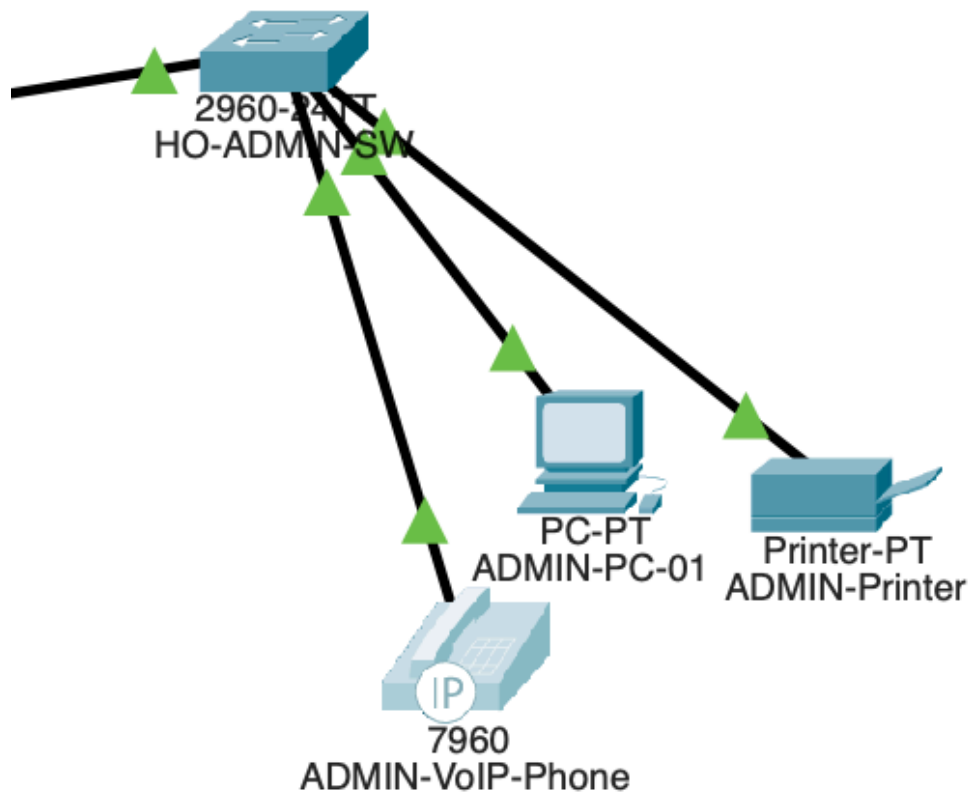


Device: HO-IT-SW

Purpose: Shows IT and Records department connectivity.

6.4 Reception/Admin Department Network

The Reception/Admin department supports administrative devices and VoIP communication.



Device: HO-ADMIN-SW

Purpose: Shows Admin department connections.

7. Branch Office Network Design

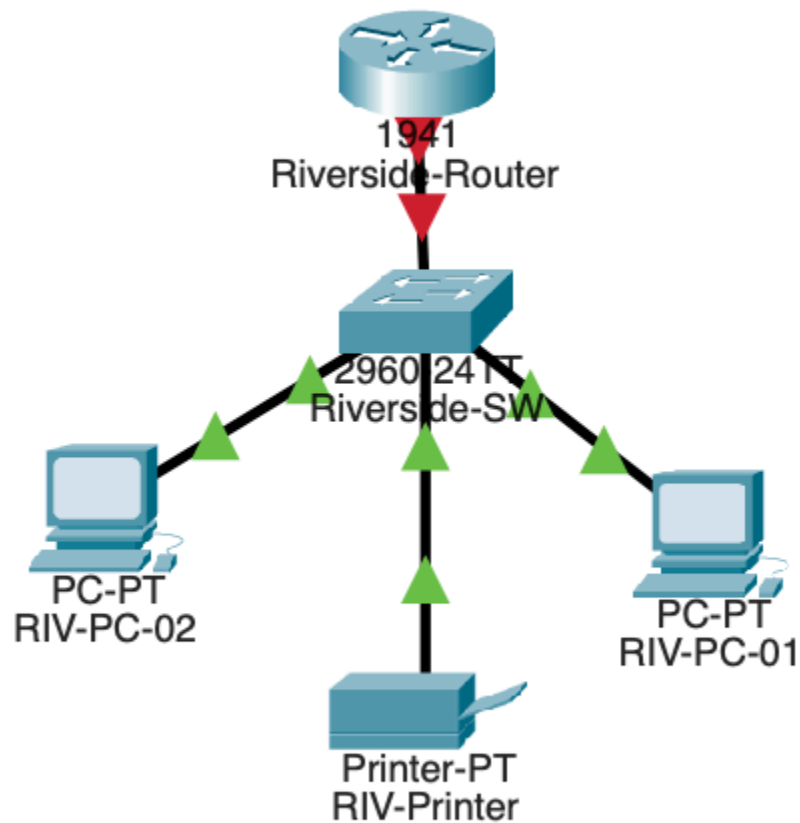
The organization contains two branch offices:

- Riverside Office
- Uptown Office

Both offices contain a router, switch, and end devices.

7.1 Riverside Office LAN

The Riverside branch follows a small star topology.

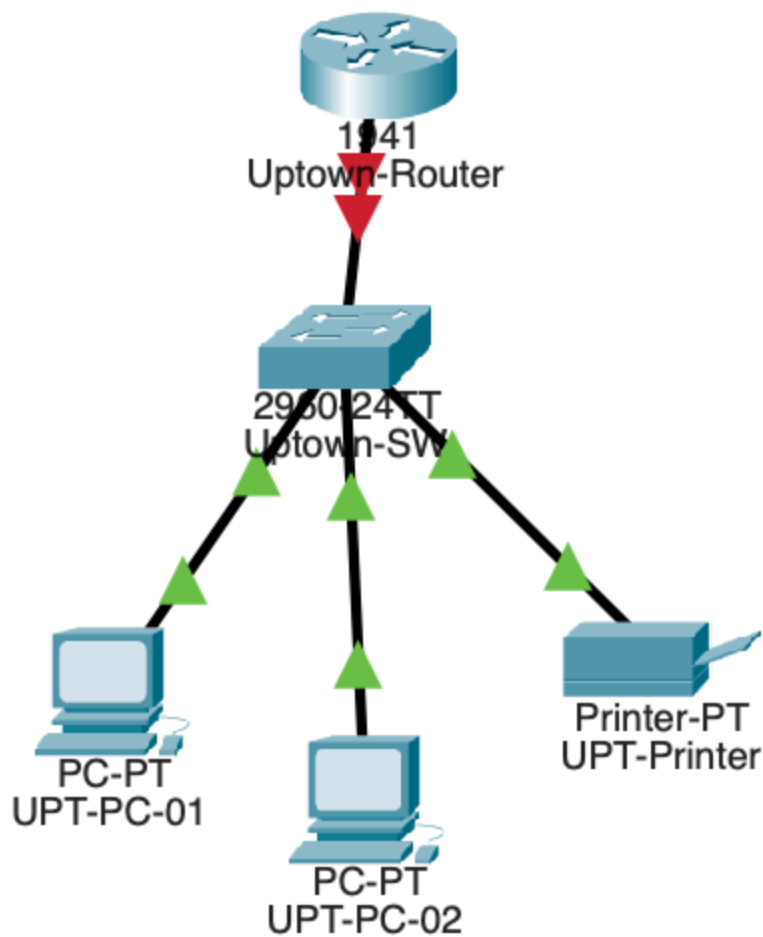


Device: Riverside Office Network

Purpose: Shows branch LAN connectivity.

7.2 Uptown Office LAN

The Uptown branch follows the same architecture.



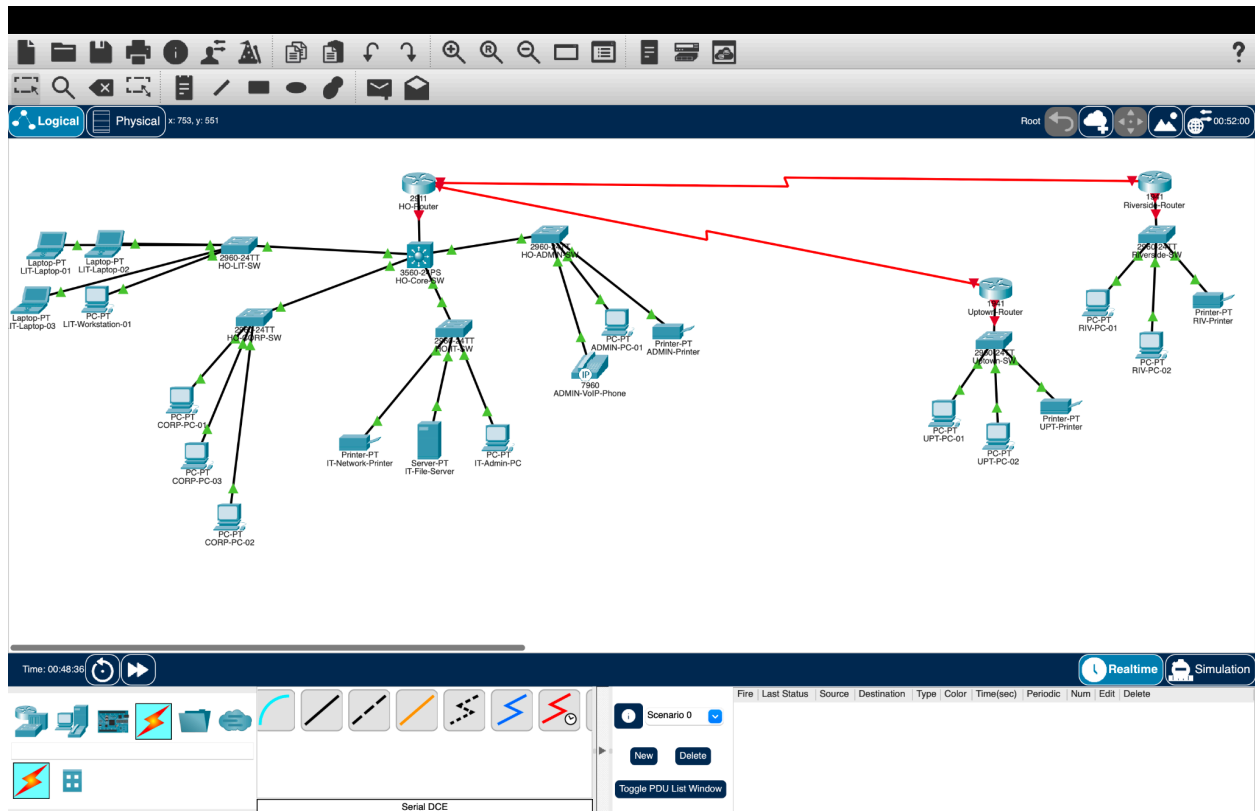
Device: Uptown Office Network

Purpose: Shows branch LAN connectivity.

8. WAN Physical Connectivity

The Head Office is connected to Riverside and Uptown offices using serial leased line connections.

These WAN links provide communication between different office locations.



Devices:

- HO-Router
- Riverside-Router
- Uptown-Router

Purpose: Shows WAN physical connectivity.

9. WAN Interface Configuration

After establishing the physical WAN connections, IP addressing was assigned to the serial interfaces of all routers.

The WAN network uses point-to-point communication between routers.

The configured WAN links are:

Connection	Network Address	Device Interface	IP Address
HO Router ↔ Riverside Router	192.168.43.0/30	HO Serial0/3/0	192.168.43.1
		Riverside Serial0/1/0	192.168.43.2
HO Router ↔ Uptown Router	192.168.43.4/30	HO Serial0/3/1	192.168.43.5
		Uptown Serial0/1/0	192.168.43.6

9.1 Head Office Router WAN Configuration

The Head Office Router was configured with serial interfaces for communication with branch routers.

Physical Config CLI Attributes

IOS Command Line Interface

```
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-Router#enable
HO-Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
HO-Router(config)#
HO-Router(config)#interface serial0/3/0
HO-Router(config-if)#ip address 192.168.43.1 255.255.255.252
HO-Router(config-if)#clock rate 64000
HO-Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
HO-Router(config-if)#exit
HO-Router(config)#
HO-Router(config)#interface serial0/3/1
HO-Router(config-if)#ip address 192.168.43.5 255.255.255.252
HO-Router(config-if)#clock rate 64000
HO-Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
HO-Router(config-if)#exit
HO-Router(config)#
HO-Router(config)#end
HO-Router#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-Router#show ip interface brief
Interface      IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0    unassigned      YES unset  administratively down  down
GigabitEthernet0/1    unassigned      YES unset  administratively down  down
GigabitEthernet0/2    unassigned      YES unset  administratively down  down
Serial0/3/0          192.168.43.1    YES manual  down                    down
Serial0/3/1          192.168.43.5    YES manual  down                    down
Vlan1               unassigned      YES unset  administratively down  down
HO-Router#
```

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Device: HO-Router

Purpose: Shows WAN interface configuration on Head Office Router.

9.2 Riverside Router WAN Configuration

The Riverside Router was configured with the WAN IP address connected to the Head Office Router.

```
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#hostname Riverside-Router
Riverside-Router(config)#enable secret Harbor@123
Riverside-Router(config)#no ip domain-lookup
Riverside-Router(config)#
Riverside-Router(config)#interface serial0/1/0
Riverside-Router(config-if)#ip address 192.168.43.2 255.255.255.252
Riverside-Router(config-if)#no shutdown

Riverside-Router(config-if)#exit
Riverside-Router(config)#
Riverside-Router(config)#end
Riverside-Router#write
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Riverside-Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

Riverside-Router#show ip interface brief
Interface      IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0    unassigned      YES unset  administratively down  down
GigabitEthernet0/1    unassigned      YES unset  administratively down  down
Serial0/1/0          192.168.43.2    YES manual    up                up
Serial0/1/1          unassigned      YES unset  administratively down  down
Vlan1              unassigned      YES unset  administratively down  down
Riverside-Router#
```

Device: Riverside-Router

Purpose: Shows Riverside WAN interface configuration.

9.3 Uptown Router WAN Configuration

The Uptown Router was configured with the WAN IP address connected to the Head Office Router.

```
Uptown-Router#show ip interface brief
Interface      IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0    unassigned      YES unset  administratively down  down
GigabitEthernet0/1    unassigned      YES unset  administratively down  down
Serial0/1/0          192.168.43.6    YES manual    up                up
Serial0/1/1          unassigned      YES unset  administratively down  down
Vlan1              unassigned      YES unset  administratively down  down
Uptown-Router#
```

Device: Uptown-Router

Purpose: Shows Uptown WAN interface configuration.

9.4 WAN Connectivity Verification

After configuring all WAN interfaces, connectivity was verified using ICMP ping testing.

Successful communication confirmed that:

- Serial links were operational.
- Routers were able to communicate.
- WAN connectivity was established.

```
HO-Router#ping 192.168.43.2
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.43.2, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/4/12 ms
```

```
HO-Router#ping 192.168.43.6
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.43.6, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/6/19 ms
```

Device: HO-Router

Purpose: Shows successful WAN connectivity testing.

10. VLAN Implementation

Virtual Local Area Networks (VLANs) were implemented to logically divide the network into separate departments.

Each VLAN represents an independent broadcast domain.

Advantages of VLAN implementation:

- Improved network security
 - Reduced broadcast traffic
 - Easier management
 - Better performance
-

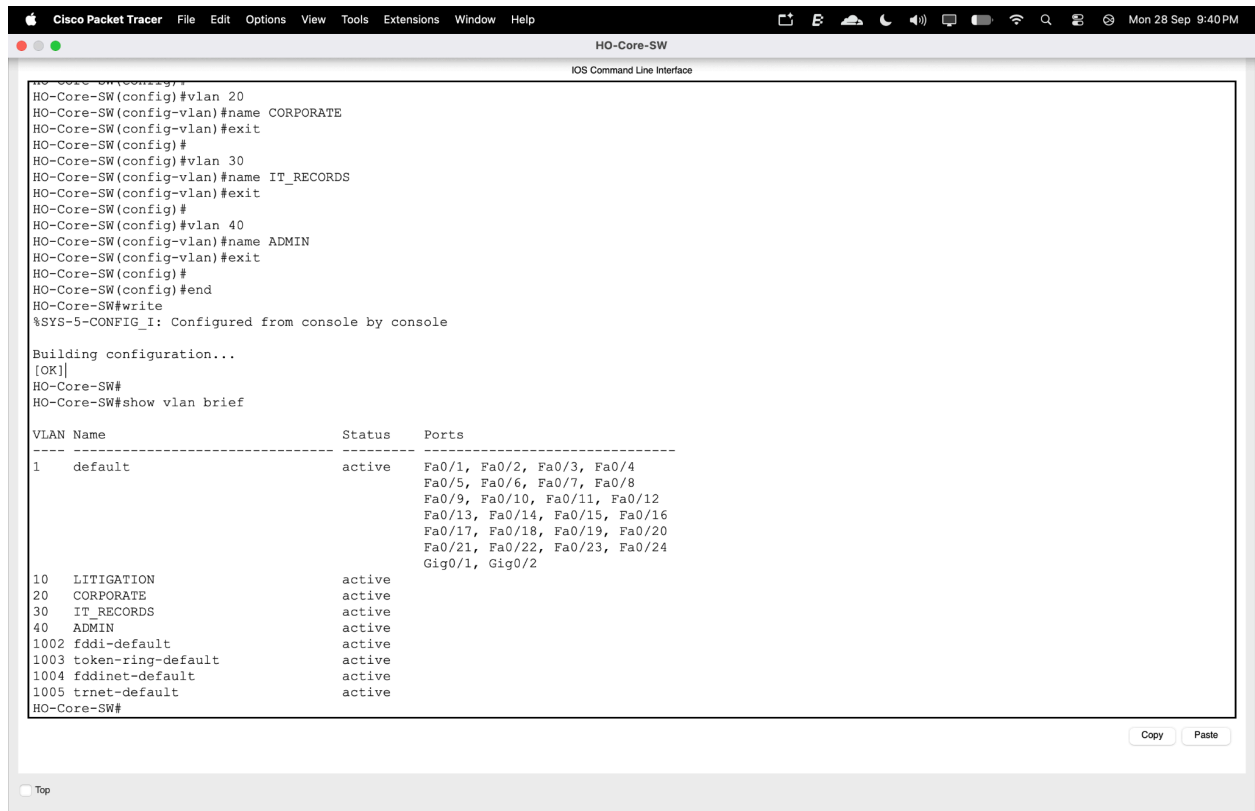
10.1 VLAN Assignment Table

VLAN ID	VLAN Name	Department
10	LITIGATION	Litigation Department
20	CORPORATE	Corporate Law Department
30	IT_RECORDS	IT & Records Department
40	ADMIN	Reception/Admin Department
50	VOICE	IP Phone Network

10.2 Core Switch VLAN Creation

All required VLANs were created on the Core Switch.

The Core Switch acts as the central VLAN management device.



```
HO-Core-SW(config)#vlan 20
HO-Core-SW(config-vlan)#name CORPORATE
HO-Core-SW(config-vlan)#exit
HO-Core-SW(config)#
HO-Core-SW(config)#vlan 30
HO-Core-SW(config-vlan)#name IT_RECORDS
HO-Core-SW(config-vlan)#exit
HO-Core-SW(config)#
HO-Core-SW(config)#vlan 40
HO-Core-SW(config-vlan)#name ADMIN
HO-Core-SW(config-vlan)#exit
HO-Core-SW(config)#
HO-Core-SW(config)#end
HO-Core-SW#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-Core-SW#
HO-Core-SW#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2

10   LITIGATION              active
20   CORPORATE               active
30   IT_RECORDS              active
40   ADMIN                   active
1002 fddi-default           active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
HO-Core-SW#
```

Device: HO-Core-SW

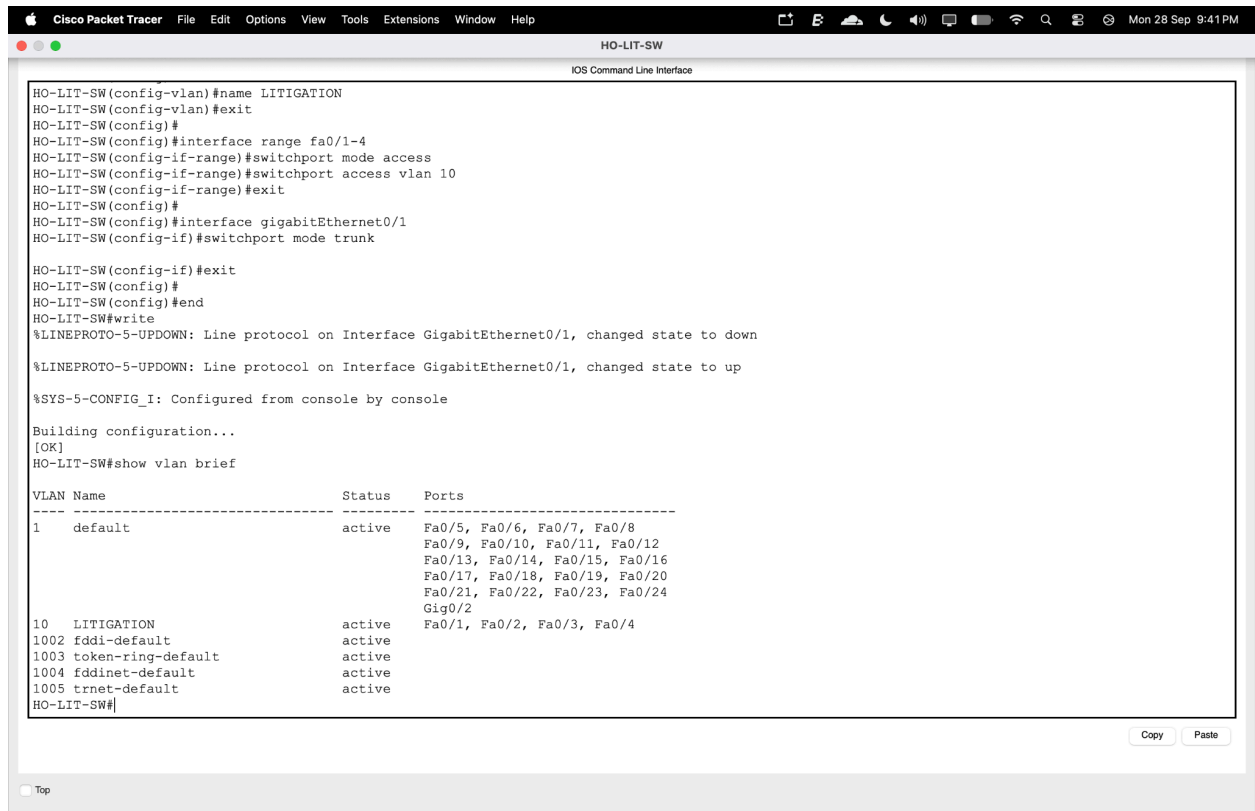
Purpose: Shows VLAN creation on Core Switch.

10.3 Department Access Switch VLAN Configuration

Each department switch was configured with the required VLAN.

Litigation Switch VLAN Configuration

The Litigation switch was assigned VLAN 10 for department devices.



```
HO-LIT-SW(config-vlan)#name LITIGATION
HO-LIT-SW(config-vlan)#exit
HO-LIT-SW(config)#
HO-LIT-SW(config)#interface range fa0/1-4
HO-LIT-SW(config-if-range)#switchport mode access
HO-LIT-SW(config-if-range)#switchport access vlan 10
HO-LIT-SW(config-if-range)#exit
HO-LIT-SW(config)#
HO-LIT-SW(config)#interface gigabitEthernet0/1
HO-LIT-SW(config-if)#switchport mode trunk

HO-LIT-SW(config-if)#exit
HO-LIT-SW(config)#
HO-LIT-SW(config)#end
HO-LIT-SW#write
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-LIT-SW#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/2
10   LITIGATION              active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
HO-LIT-SW#
```

Device: HO-LIT-SW

Purpose: Shows Litigation VLAN configuration.

Corporate Law Switch VLAN Configuration

The Corporate Law switch was assigned VLAN 20.


```
HO-CORP-SW(config-if-range Corporate)
HO-CORP-SW(config-vlan)#exit
HO-CORP-SW(config)#
HO-CORP-SW(config)#interface range fa0/1-3
HO-CORP-SW(config-if-range)#switchport mode access
HO-CORP-SW(config-if-range)#switchport access vlan 20
HO-CORP-SW(config-if-range)#exit
HO-CORP-SW(config)#
HO-CORP-SW(config)#interface gigabitEthernet0/1
HO-CORP-SW(config-if)#switchport mode trunk

HO-CORP-SW(config-if)#exit
HO-CORP-SW(config)#
HO-CORP-SW(config)#end
HO-CORP-SW#write
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-CORP-SW#
HO-CORP-SW#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                           Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24, Gig0/2
20   CORPORATE              active    Fa0/1, Fa0/2, Fa0/3
1002 fddi-default          active
1003 token-ring-default   active
1004 fddinet-default       active
1005 trnet-default        active
HO-CORP-SW#
```

Device: HO-CORP-SW

Purpose: Shows Corporate VLAN configuration.

IT & Records Switch VLAN Configuration

The IT switch was assigned VLAN 30.

```
HO-IT-SW(config-vlan)#exit
HO-IT-SW(config)#
HO-IT-SW(config)#interface range fa0/1-3
HO-IT-SW(config-if-range)#switchport mode access
HO-IT-SW(config-if-range)#switchport access vlan 30
HO-IT-SW(config-if-range)#exit
HO-IT-SW(config)#
HO-IT-SW(config)#interface gigabitEthernet0/1
HO-IT-SW(config-if)#switchport mode trunk

HO-IT-SW(config-if)#exit
HO-IT-SW(config)#
HO-IT-SW(config)#end
HO-IT-SW#write
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-IT-SW#
HO-IT-SW#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                           Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24, Gig0/2
30   IT_RECORDS              active    Fa0/1, Fa0/2, Fa0/3
1002 fddi-default            active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default         active
HO-IT-SW#
```

Device: HO-IT-SW

Purpose: Shows IT Records VLAN configuration.

Admin Switch VLAN Configuration

The Admin switch was assigned VLAN 40.

```
HO-ADMIN-SW(config-vlan)#exit
HO-ADMIN-SW(config)#
HO-ADMIN-SW(config)#interface range fa0/1-3
HO-ADMIN-SW(config-if-range)#switchport mode access
HO-ADMIN-SW(config-if-range)#switchport access vlan 40
HO-ADMIN-SW(config-if-range)#exit
HO-ADMIN-SW(config)#
HO-ADMIN-SW(config)#interface gigabitEthernet0/1
HO-ADMIN-SW(config-if)#switchport mode trunk

HO-ADMIN-SW(config-if)#exit
HO-ADMIN-SW(config)#
HO-ADMIN-SW(config)#end
HO-ADMIN-SW#write
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-ADMIN-SW#
HO-ADMIN-SW#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/2
40	ADMIN	active	Fa0/1, Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Device: HO-ADMIN-SW

Purpose: Shows Admin VLAN configuration.

11. Inter-VLAN Routing Implementation

Since VLANs create separate broadcast domains, communication between VLANs requires Layer 3 routing.

Router-on-a-Stick methodology was implemented.

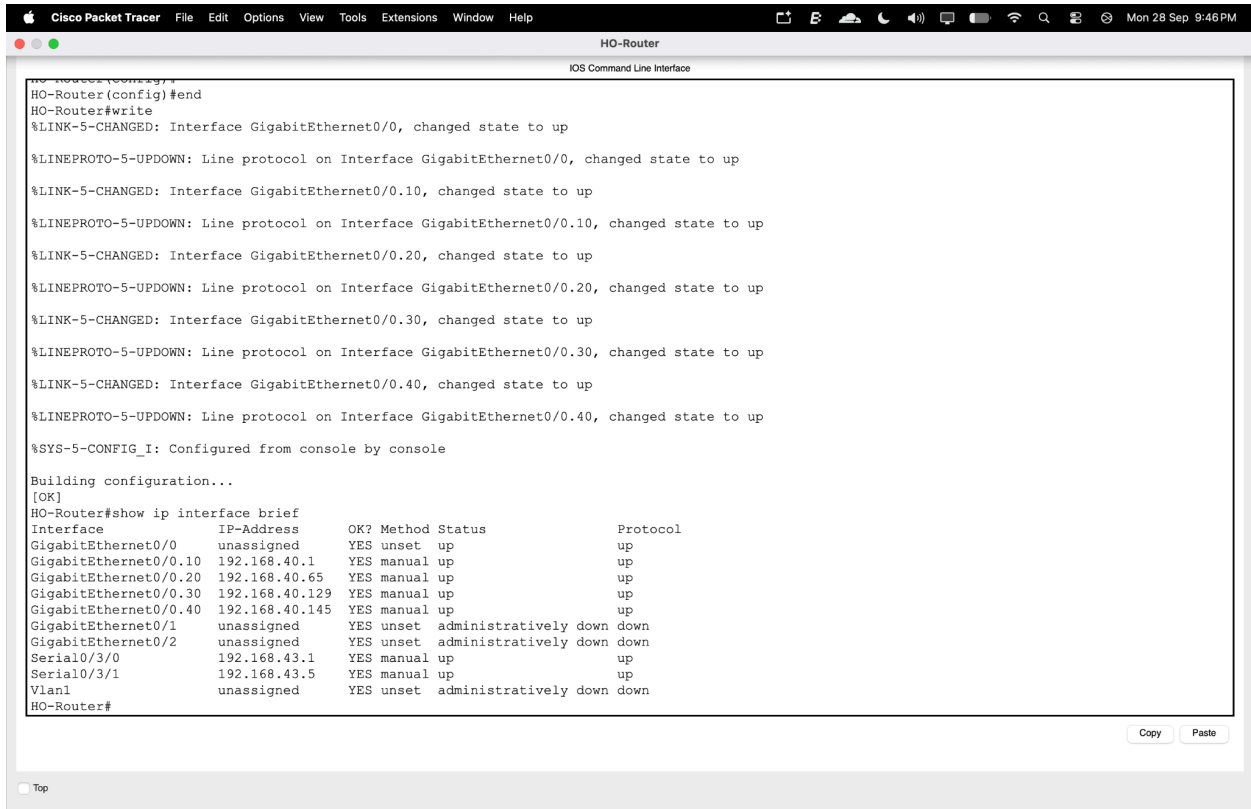
In this method:

- One physical router interface is connected to the Core Switch.
- Multiple logical subinterfaces are created.
- Each subinterface acts as a gateway for a VLAN.

11.1 Router Subinterface Configuration

The Head Office Router was configured with the following gateways:

VLAN	Interface	Gateway Address
VLAN 10	G0/0.10	192.168.40.1
VLAN 20	G0/0.20	192.168.40.65
VLAN 30	G0/0.30	192.168.40.129
VLAN 40	G0/0.40	192.168.40.145
VLAN 50	G0/0.50	192.168.40.161



Device: HO-Router

Purpose: Shows Router-on-a-Stick configuration.

12. Trunk Configuration

Trunk links were configured between:

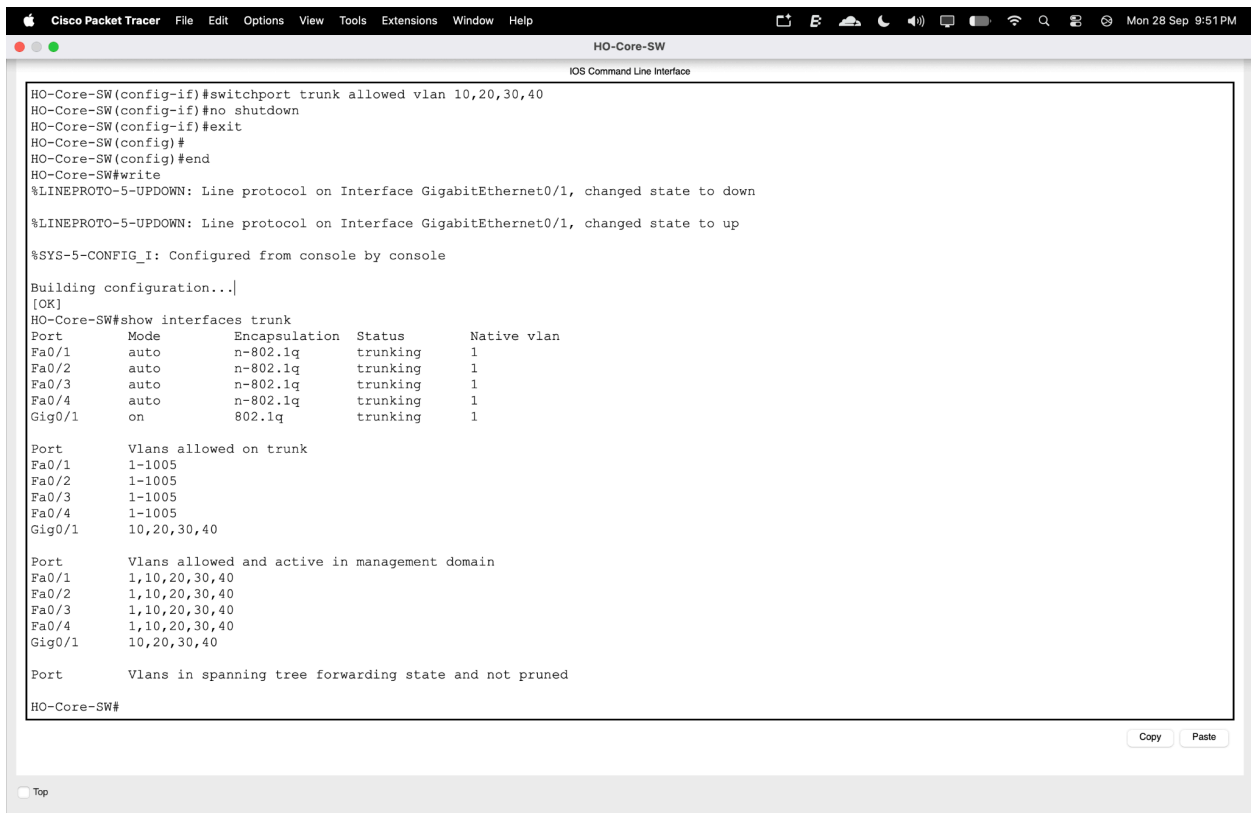
- Core Switch and Router
- Core Switch and Department Switches

Trunking allows multiple VLAN traffic to travel through a single physical connection using IEEE 802.1Q tagging.

Allowed VLANs:

- VLAN 10
- VLAN 20
- VLAN 30
- VLAN 40

- VLAN 50



```
HO-Core-SW(config-if)#switchport trunk allowed vlan 10,20,30,40
HO-Core-SW(config-if)#no shutdown
HO-Core-SW(config-if)#exit
HO-Core-SW(config)#
HO-Core-SW(config)#end
HO-Core-SW#write
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-Core-SW#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     auto      n-802.1q       trunking    1
Fa0/2     auto      n-802.1q       trunking    1
Fa0/3     auto      n-802.1q       trunking    1
Fa0/4     auto      n-802.1q       trunking    1
Gig0/1    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/2     1-1005
Fa0/3     1-1005
Fa0/4     1-1005
Gig0/1    10,20,30,40

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,40
Fa0/2     1,10,20,30,40
Fa0/3     1,10,20,30,40
Fa0/4     1,10,20,30,40
Gig0/1    10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30,40
Fa0/2     1,10,20,30,40
Fa0/3     1,10,20,30,40
Fa0/4     1,10,20,30,40
Gig0/1    10,20,30,40

HO-Core-SW#
```

Device: HO-Core-SW

Purpose: Shows trunk configuration and allowed VLANs.

13. IP Addressing Design

A VLSM-based addressing scheme was implemented to efficiently allocate IP addresses.

The main private network used:

192.168.40.0/24

This network was divided according to department requirements.

13.1 VLSM Address Allocation

Department	Network Address	Subnet Mask	Gateway
Litigation VLAN 10	192.168.40.0/26	255.255.255.192	192.168.40.1
Corporate VLAN 20	192.168.40.64/26	255.255.255.192	192.168.40.65
IT Records VLAN 30	192.168.40.128/28	255.255.255.240	192.168.40.129
Admin VLAN 40	192.168.40.144/28	255.255.255.240	192.168.40.145
Voice VLAN 50	192.168.40.160/28	255.255.255.240	192.168.40.161

14. End Device IP Configuration

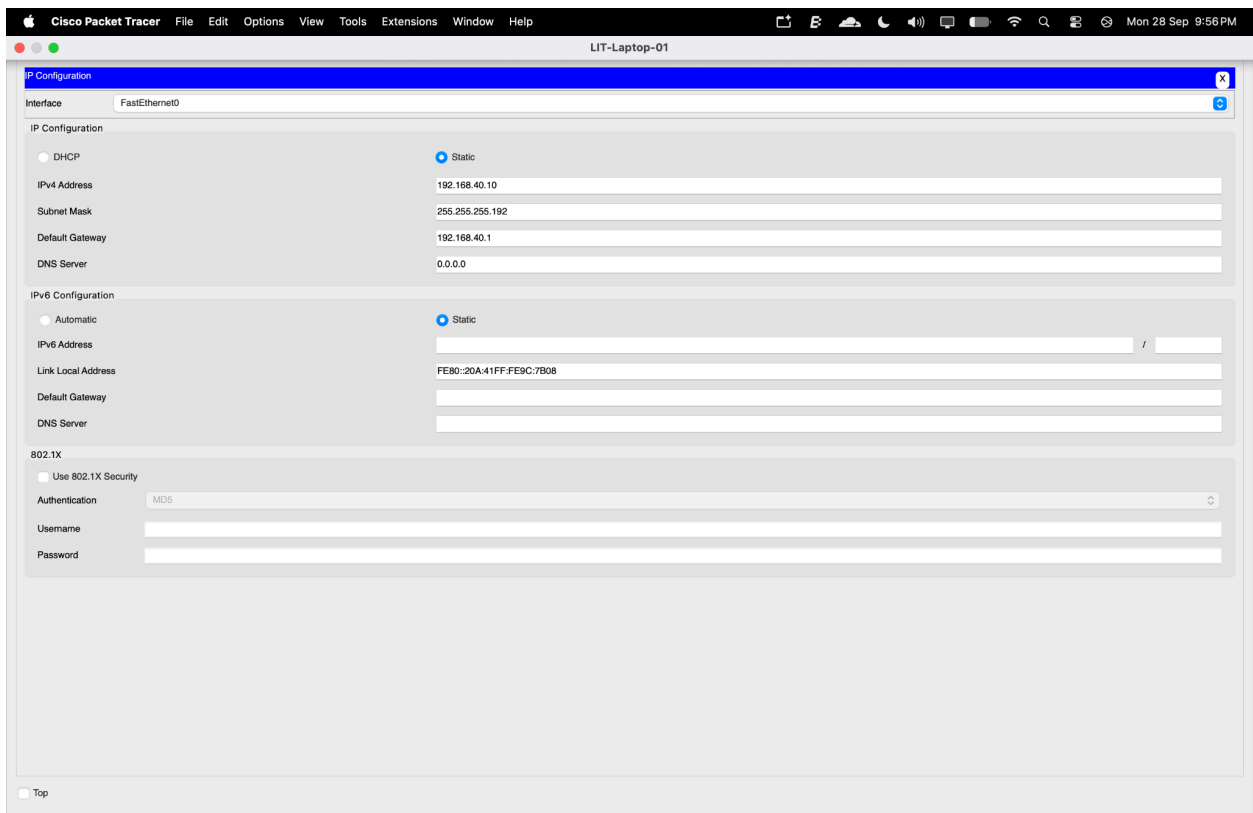
Each department device was configured according to its assigned VLAN network.

14.1 Litigation Device Configuration

Devices in the Litigation department were assigned addresses from VLAN 10.

Example:

Device	IP Address	Gateway
Litigation Laptop	192.168.40.x	192.168.40.1

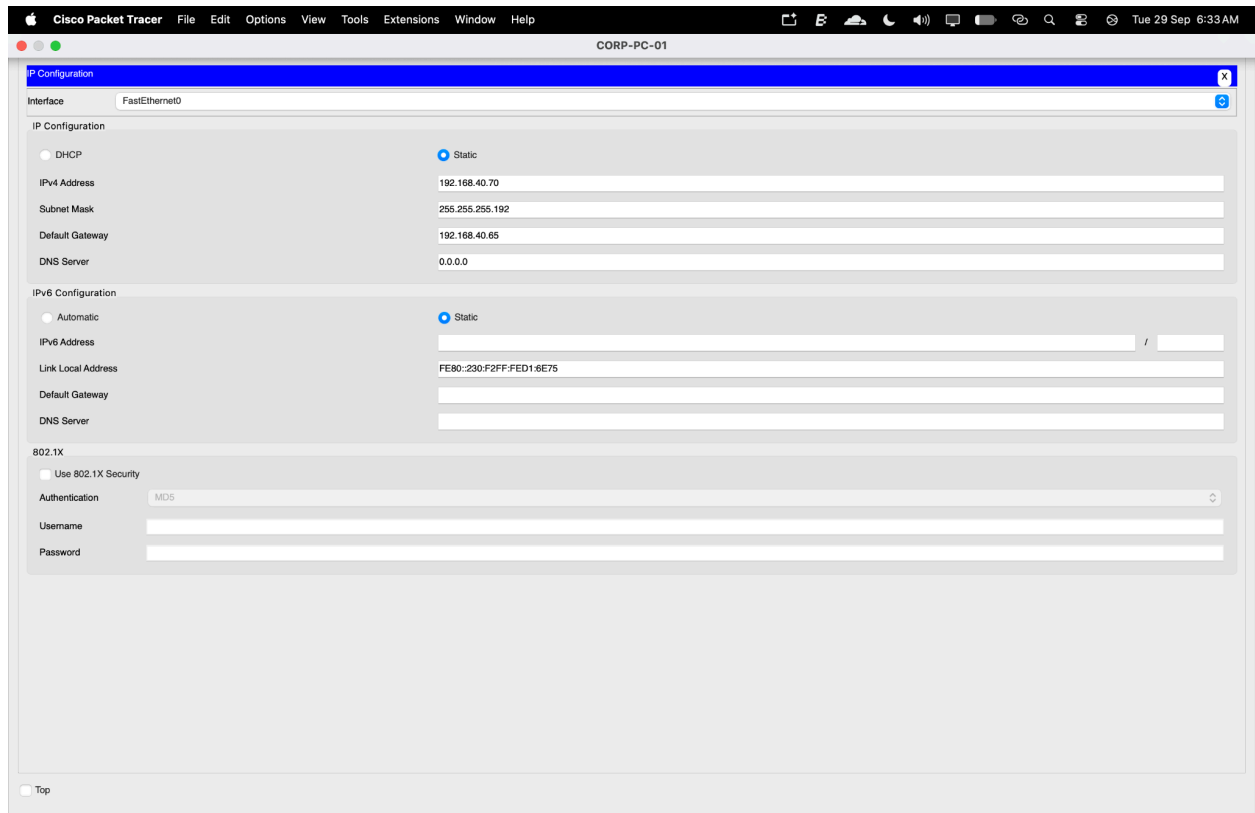


Device: Litigation Laptop

Purpose: Shows Litigation device IP configuration.

14.2 Corporate Device Configuration

Corporate devices were assigned addresses from VLAN 20.

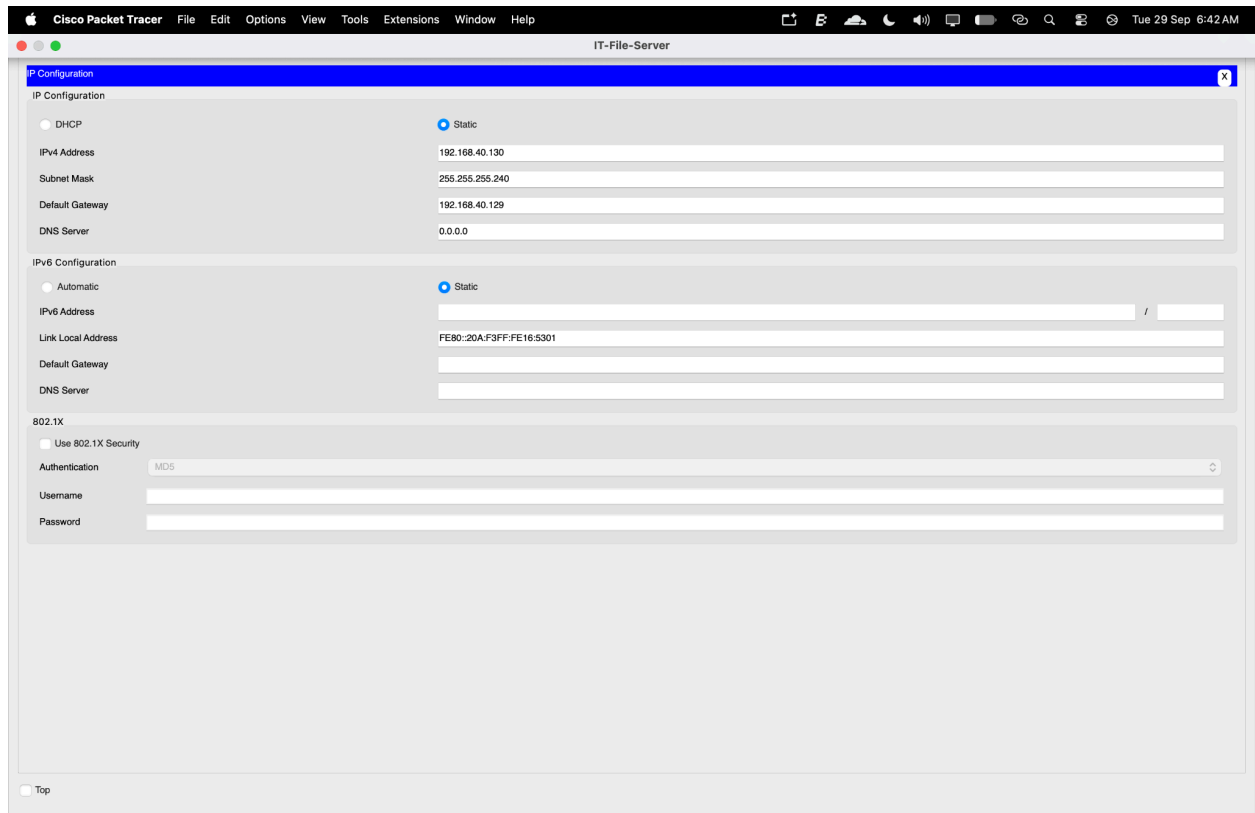


Device: Corporate Device

Purpose: Shows Corporate device IP configuration.

14.3 IT Records Device Configuration

IT devices were configured in VLAN 30.

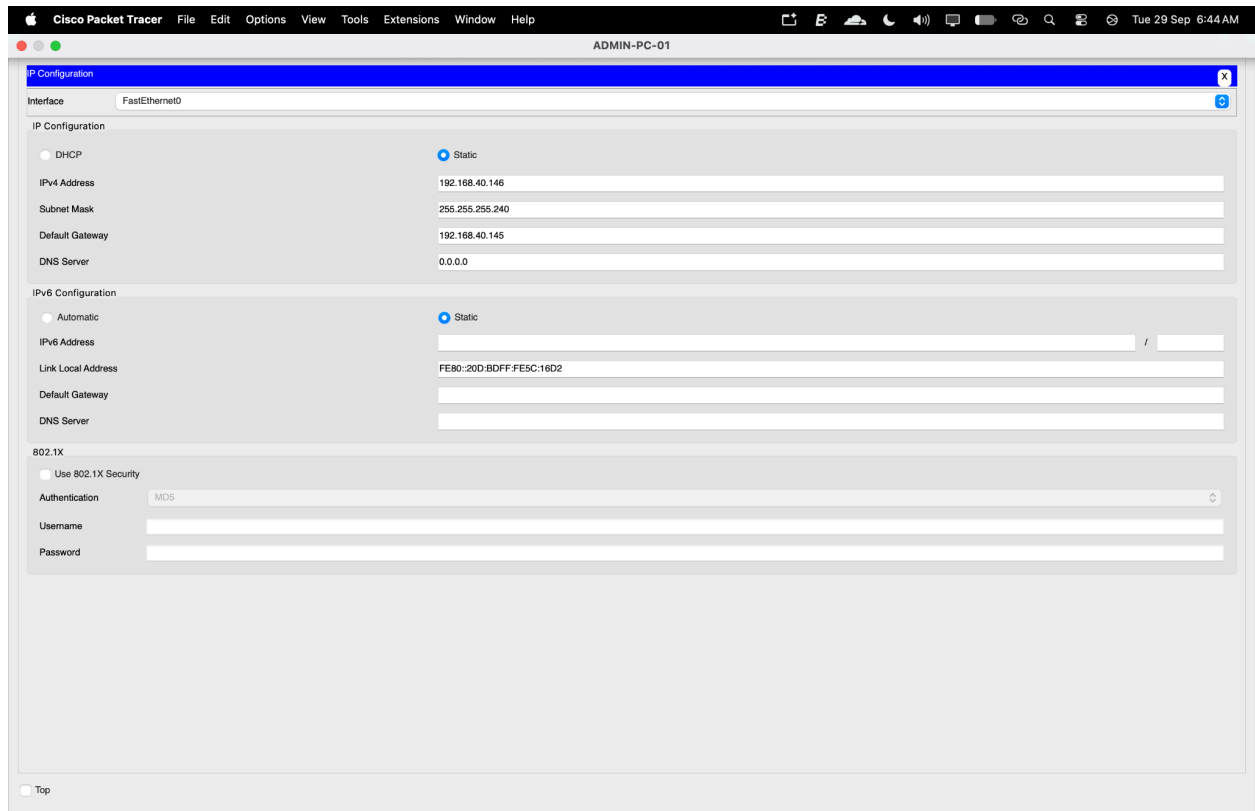


Device: IT Records Device

Purpose: Shows IT device IP configuration.

14.4 Admin Device Configuration

Administrative devices were configured in VLAN 40.



Device: Admin Device

Purpose: Shows Admin device IP configuration.

15. Voice VLAN Implementation

Voice communication is an important requirement for enterprise networks. A separate Voice VLAN was implemented to isolate voice traffic from normal data traffic.

The advantages of Voice VLAN are:

- Better voice quality
- Reduced congestion
- Improved security
- Separate management of IP phones

The Voice VLAN used in this network is:

VLAN ID	VLAN Name	Purpose
50	VOICE	IP Phone Communication

15.1 Voice VLAN Configuration

The Admin switch port connected to the IP Phone was configured with:

- Data VLAN: VLAN 40 (ADMIN)
- Voice VLAN: VLAN 50 (VOICE)

This allows the same physical port to carry both:

- Computer traffic
 - IP Phone traffic
-

[Insert Screenshot: SS27_Voice_VLAN_Configuration.png]

Device: HO-ADMIN-SW

Purpose: Shows Voice VLAN configuration on the switch port.

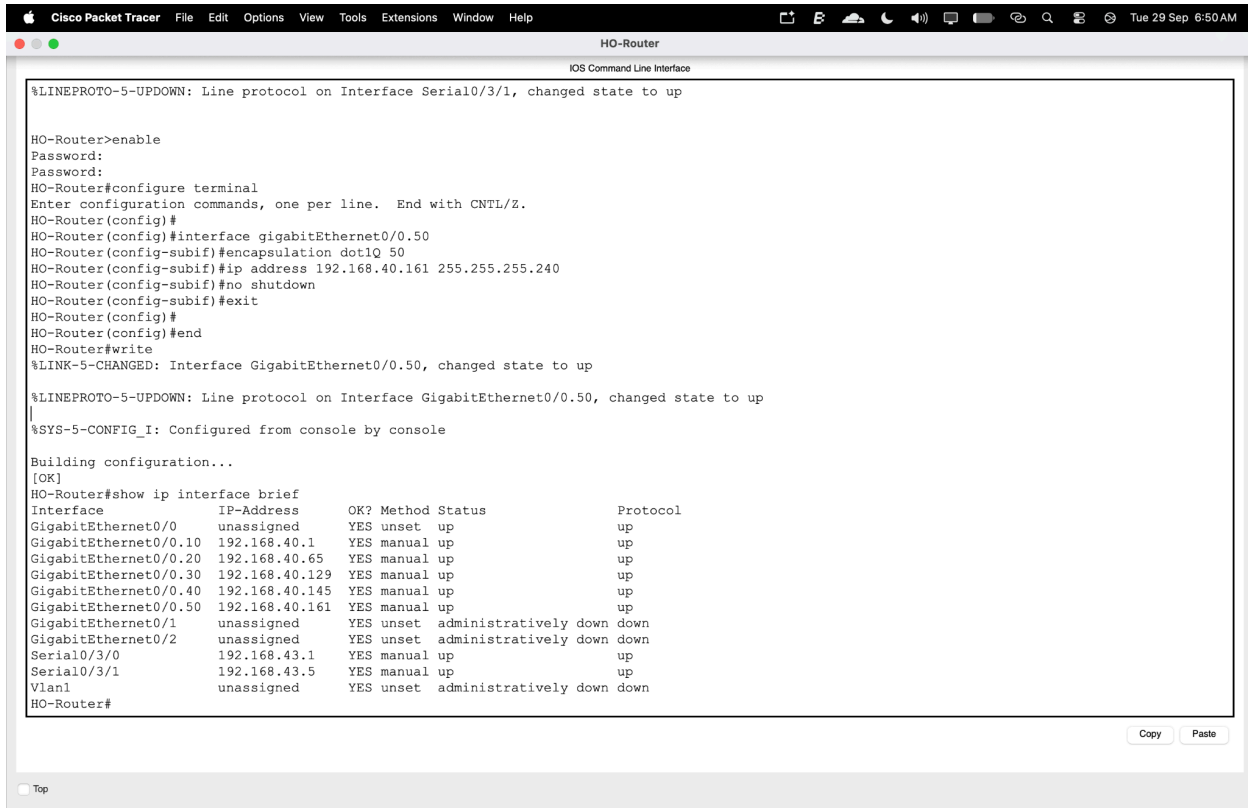
15.2 Voice VLAN Router Subinterface

The Head Office Router was configured with a dedicated subinterface for Voice VLAN.

The gateway assigned to the Voice VLAN is:

VLAN	Interface	Gateway
------	-----------	---------

VLAN 50 G0/0.50 192.168.40.16
1



```
HO-Router>enable
Password:
Password:
HO-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
HO-Router(config)#
HO-Router(config)#interface gigabitEthernet0/0.50
HO-Router(config-subif)#encapsulation dot1Q 50
HO-Router(config-subif)#ip address 192.168.40.161 255.255.255.240
HO-Router(config-subif)#no shutdown
HO-Router(config-subif)#exit
HO-Router(config)#
HO-Router(config)#end
HO-Router#write
%LINK-5-CHANGED: Interface GigabitEthernet0/0.50, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.50, changed state to up
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0      unassigned      YES unset  up          up
GigabitEthernet0/0.10   192.168.40.1    YES manual  up          up
GigabitEthernet0/0.20   192.168.40.65   YES manual  up          up
GigabitEthernet0/0.30   192.168.40.129  YES manual  up          up
GigabitEthernet0/0.40   192.168.40.145  YES manual  up          up
GigabitEthernet0/0.50   192.168.40.161  YES manual  up          up
GigabitEthernet0/1      unassigned      YES unset  administratively down down
GigabitEthernet0/2      unassigned      YES unset  administratively down down
Serial0/3/0            192.168.43.1    YES manual  up          up
Serial0/3/1            192.168.43.5    YES manual  up          up
Vlan1                  unassigned      YES unset  administratively down down
HO-Router#
```

Device: HO-Router

Purpose: Shows Voice VLAN gateway configuration.

15.3 DHCP Configuration for IP Phone

A DHCP pool was created to automatically assign IP addresses to IP phones.

DHCP Pool Details:

Parameter

Value

Pool Name VOICE_POOL

Network 192.168.40.160/28

Default Gateway 192.168.40.161

Address Range 192.168.40.161 –
 192.168.40.174

```
HO-Router#show ip dhcp pool
```

```
Pool VOICE_POOL :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)          : 0 / 0
  Total addresses                   : 14
  Leased addresses                  : 0
  Excluded addresses                : 1
  Pending event                    : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  192.168.40.161    192.168.40.161 - 192.168.40.174    0 / 1 / 14
HO-Router#
```

Device: HO-Router

Purpose: Shows DHCP pool configuration for Voice VLAN.

15.4 IP Phone DHCP Assignment

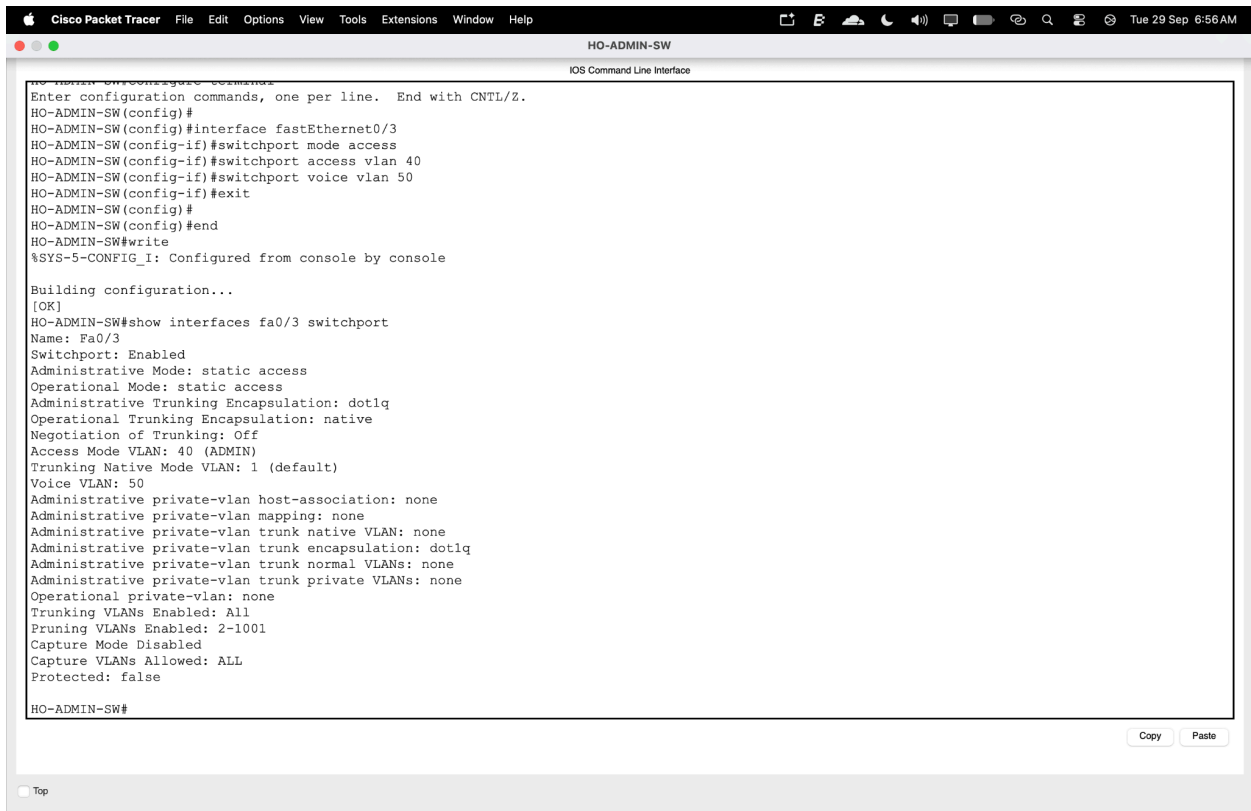
After connecting the IP Phone to the configured switch port, the device automatically received an IP address from the DHCP server.

The IP Phone received:

Device **IP Address**

IP 192.168.40.16

Phone 2



The screenshot shows the Cisco Packet Tracer interface with a terminal window open for the device 'HO-ADMIN-SW'. The terminal displays the following commands and output:

```
HO-ADMIN-SW#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
HO-ADMIN-SW(config)#
HO-ADMIN-SW(config)#interface fastEthernet0/3
HO-ADMIN-SW(config-if)#switchport mode access
HO-ADMIN-SW(config-if)#switchport access vlan 40
HO-ADMIN-SW(config-if)#switchport voice vlan 50
HO-ADMIN-SW(config-if)#exit
HO-ADMIN-SW(config)#
HO-ADMIN-SW(config)#end
HO-ADMIN-SW#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
HO-ADMIN-SW#show interfaces fa0/3 switchport
Name: Fa0/3
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 40 (ADMIN)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: 50
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false

HO-ADMIN-SW#
```

Device: IP Phone

Purpose: Shows successful IP address assignment.

15.5 DHCP Binding Verification

The DHCP binding table was checked on the router to verify the assigned IP address.

The router successfully displayed the IP Phone MAC address and assigned IP address.

```
HO-Router>show ip dhcp binding
IP address      Client-ID/      Lease expiration  Type
                Hardware address
192.168.40.162  00D0.5879.00C9  --               Automatic
HO-Router>
```

Device: HO-Router

Purpose: Confirms DHCP lease allocation.

16. Network Testing and Verification

After completing the configuration, different tests were performed to verify proper network operation.

The testing included:

- Gateway connectivity testing
 - Inter-VLAN communication testing
 - WAN connectivity testing
 - Routing verification
-

16.1 Default Gateway Testing

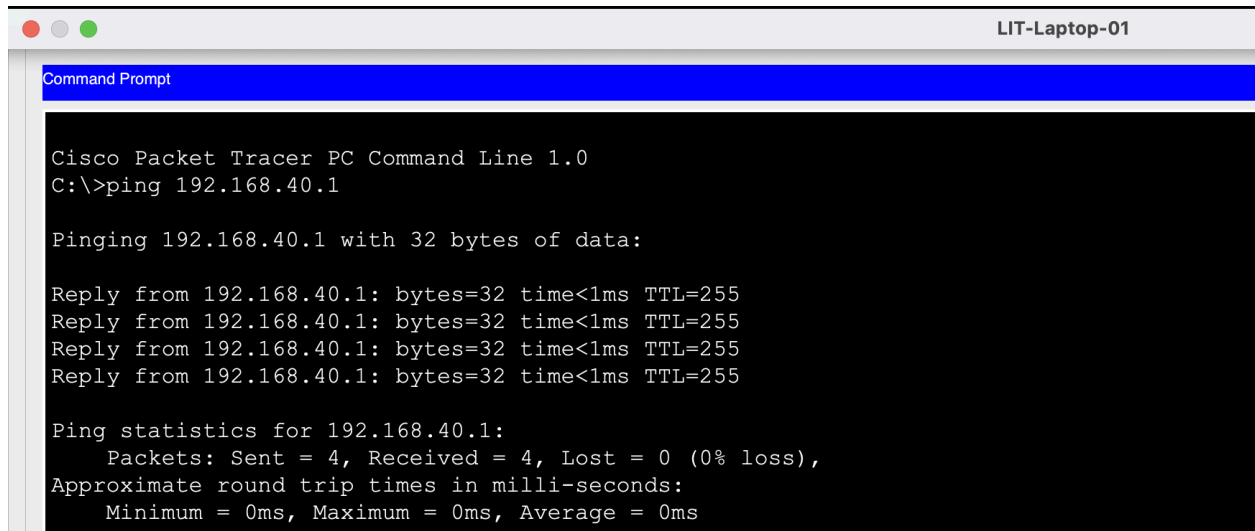
Each VLAN device was tested against its respective default gateway.

Litigation VLAN Gateway Test

The Litigation laptop successfully communicated with the VLAN 10 gateway.

Gateway:

192.168.40.1



```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Device: Litigation Laptop

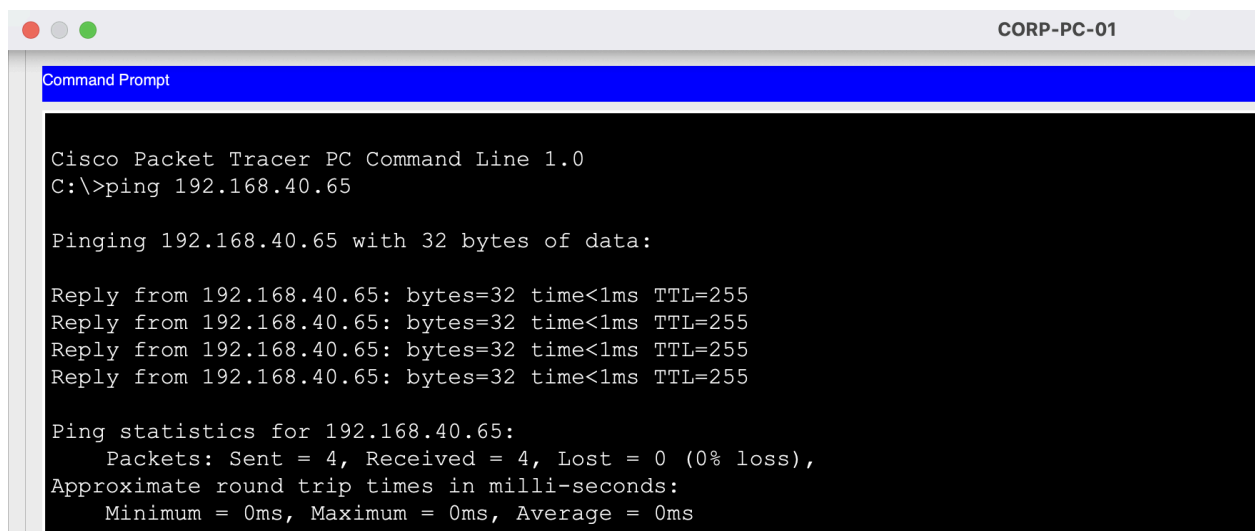
Purpose: Verifies VLAN 10 gateway connectivity.

Corporate VLAN Gateway Test

The Corporate device successfully communicated with the VLAN 20 gateway.

Gateway:

192.168.40.65



```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.65

Pinging 192.168.40.65 with 32 bytes of data:

Reply from 192.168.40.65: bytes=32 time<1ms TTL=255
Reply from 192.168.40.65: bytes=32 time<1ms TTL=255
Reply from 192.168.40.65: bytes=32 time<1ms TTL=255
Reply from 192.168.40.65: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Device: Corporate Device

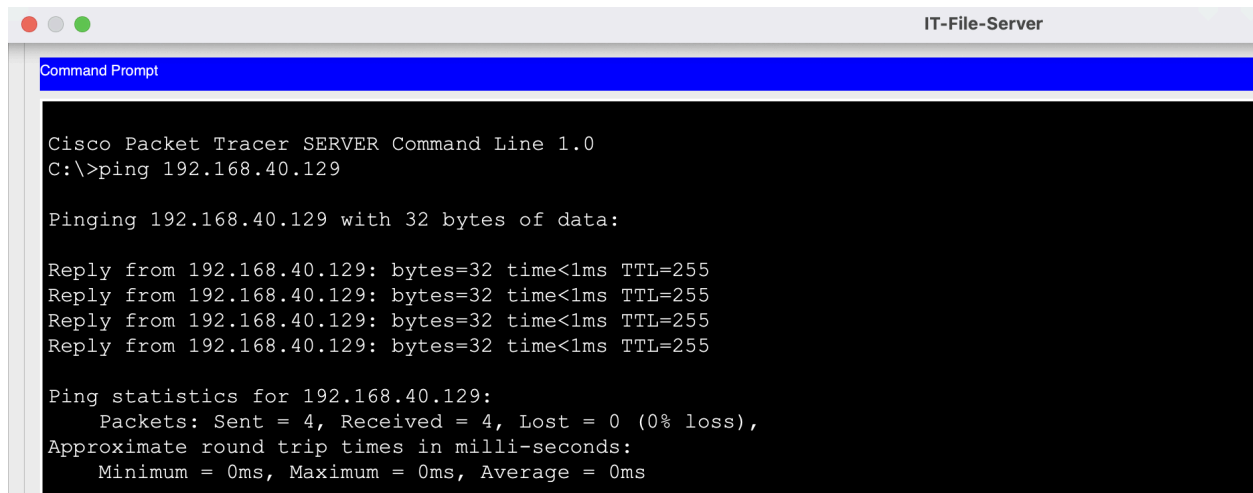
Purpose: Verifies VLAN 20 gateway connectivity.

IT Records VLAN Gateway Test

The IT device successfully communicated with the VLAN 30 gateway.

Gateway:

192.168.40.129



```
IT-File-Server
Command Prompt

Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.40.129

Pinging 192.168.40.129 with 32 bytes of data:

Reply from 192.168.40.129: bytes=32 time<1ms TTL=255
Reply from 192.168.40.129: bytes=32 time<1ms TTL=255
Reply from 192.168.40.129: bytes=32 time<1ms TTL=255
Reply from 192.168.40.129: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Device: IT Records Device

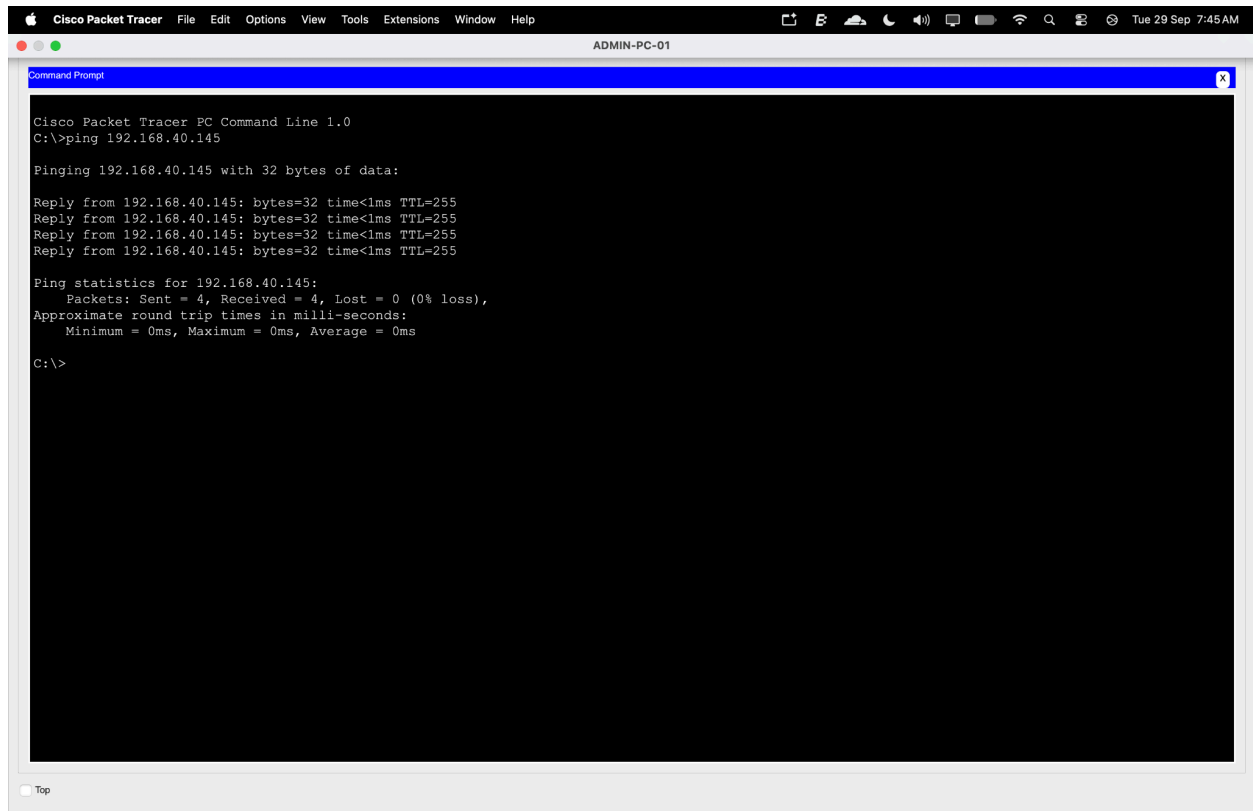
Purpose: Verifies VLAN 30 gateway connectivity.

Admin VLAN Gateway Test

The Admin device successfully communicated with the VLAN 40 gateway.

Gateway:

192.168.40.145



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.145

Pinging 192.168.40.145 with 32 bytes of data:

Reply from 192.168.40.145: bytes=32 time<1ms TTL=255
Reply from 192.168.40.145: bytes=32 time<1ms TTL=255
Reply from 192.168.40.145: bytes=32 time<1ms TTL=255
Reply from 192.168.40.145: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.145:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Device: Admin Device

Purpose: Verifies VLAN 40 gateway connectivity.

16.2 Inter-VLAN Communication Testing

Since all VLANs are separated logically, communication between them requires router-based routing.

The following tests were performed.

Litigation VLAN to IT VLAN Communication

A device from VLAN 10 successfully communicated with a device in VLAN 30.

This confirms:

- Router-on-a-Stick operation

- Correct gateway configuration
 - Proper routing between VLANs
-

```
C:\>ping 192.168.40.130

Pinging 192.168.40.130 with 32 bytes of data:

Reply from 192.168.40.130: bytes=32 time<1ms TTL=127
Reply from 192.168.40.130: bytes=32 time<1ms TTL=127
Reply from 192.168.40.130: bytes=32 time<1ms TTL=127
Reply from 192.168.40.130: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Source: Litigation VLAN

Destination: IT Records VLAN

Admin VLAN to Corporate VLAN Communication

A device from VLAN 40 successfully communicated with a device in VLAN 20.

```
C:\>ping 192.168.40.70

Pinging 192.168.40.70 with 32 bytes of data:

Reply from 192.168.40.70: bytes=32 time<1ms TTL=127
Reply from 192.168.40.70: bytes=32 time=1ms TTL=127
Reply from 192.168.40.70: bytes=32 time<1ms TTL=127
Reply from 192.168.40.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Source: Admin VLAN

Destination: Corporate VLAN

16.3 WAN Connectivity Testing

Communication between different offices was tested through the configured serial WAN links.

Head Office to Riverside Communication

The Head Office Router successfully communicated with Riverside Router.

WAN Link:

192.168.43.1 ↔ 192.168.43.2

```
HO-Router>ping 192.168.43.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.43.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/4/15 ms
```

Device: HO-Router

Purpose: Verifies WAN connectivity with Riverside office.

Head Office to Uptown Communication

The Head Office Router successfully communicated with Uptown Router.

WAN Link:

192.168.43.5 ↔ 192.168.43.6

```
HO-Router>ping 192.168.43.6
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.43.6, timeout is 2 seconds:
```

```
!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/5/13 ms
```

Device: HO-Router

Purpose: Verifies WAN connectivity with Uptown office.

16.4 Routing Table Verification

The routing table of the Head Office Router was checked to verify all connected networks.

The routing table confirmed:

- VLAN networks
 - WAN networks
 - Connected interfaces
-

```
HO-Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
      192.168.40.0/24 is variably subnetted, 10 subnets, 3 masks
C       192.168.40.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.40.64/26 is directly connected, GigabitEthernet0/0.20
L       192.168.40.65/32 is directly connected, GigabitEthernet0/0.20
C       192.168.40.128/28 is directly connected, GigabitEthernet0/0.30
L       192.168.40.129/32 is directly connected, GigabitEthernet0/0.30
C       192.168.40.144/28 is directly connected, GigabitEthernet0/0.40
L       192.168.40.145/32 is directly connected, GigabitEthernet0/0.40
C       192.168.40.160/28 is directly connected, GigabitEthernet0/0.50
L       192.168.40.161/32 is directly connected, GigabitEthernet0/0.50
      192.168.43.0/24 is variably subnetted, 4 subnets, 2 masks
```

Device: HO-Router

Purpose: Shows routing table verification.

17. Network Protocol Explanation

17.1 DHCP Process

Dynamic Host Configuration Protocol (DHCP) automatically provides network configuration to devices.

The DHCP process follows:

1. DHCP Discover

The device broadcasts a request asking for an available DHCP server.

2. DHCP Offer

The DHCP server offers an available IP address.

3. DHCP Request

The device requests the offered IP address.

4. DHCP Acknowledgement

The DHCP server confirms the assignment.

In this project, DHCP was used for automatic IP assignment of the IP Phone.

17.2 ARP Process

Address Resolution Protocol (ARP) converts IP addresses into MAC addresses.

Example:

When a device wants to communicate with another device in the same network:

1. It checks its ARP table.
 2. If MAC address is unavailable, it broadcasts an ARP request.
 3. The destination device replies with its MAC address.
 4. Communication begins.
-

17.3 DNS Process

Domain Name System (DNS) converts domain names into IP addresses.

Example:

www.example.com



IP Address

DNS makes network communication easier because users do not need to remember numerical IP addresses.

17.4 ICMP Process

Internet Control Message Protocol (ICMP) is used for network testing.

The ping command uses ICMP messages:

- Echo Request
- Echo Reply

Successful ping results confirm connectivity between devices.

18. OSI Model Mapping

OSI Layer	Implementation in Project
Layer 7 Application	DHCP, DNS services
Layer 6 Presentation	Data formatting
Layer 5 Session	Communication sessions
Layer 4 Transport	TCP/UDP communication
Layer 3 Network	IP addressing, routing, WAN communication
Layer 2 Data Link	VLANs, MAC addressing, trunking
Layer 1 Physical	Cables, interfaces, serial links

19. Private and Public IP Addressing

The network uses private IPv4 addressing.

Private IP addresses are used inside organizations and are not directly reachable from the public internet.

The implemented private network:

192.168.40.0/24

Advantages:

- Conserves public IP addresses
- Improves internal security
- Allows internal network expansion

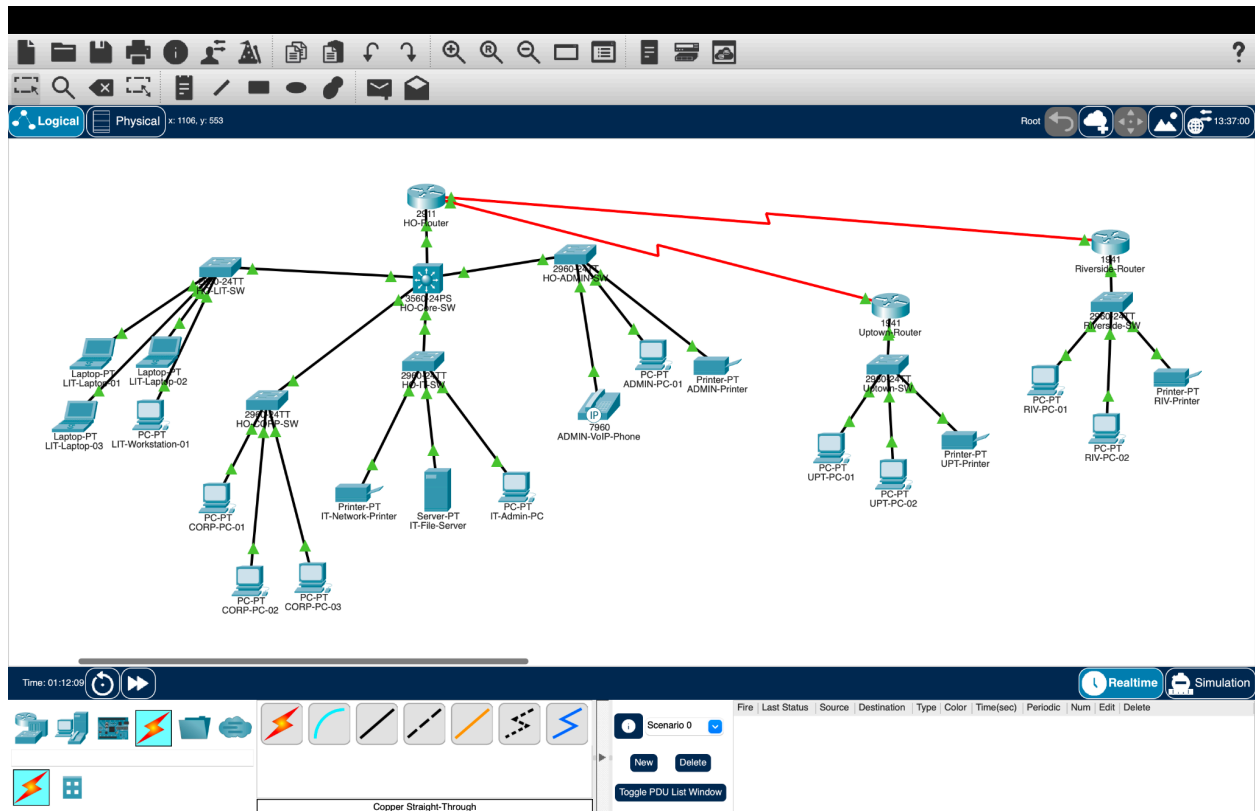
Public IP addresses are globally unique addresses used for internet communication.

20. Final Network Topology

After completing all configurations and verification, the final network topology was captured.

The final topology includes:

- Head Office network
 - Riverside branch
 - Uptown branch
 - VLAN segmentation
 - WAN connectivity
 - End devices
-



Device: Complete Packet Tracer Network

Purpose: Final implementation overview.

21. Conclusion

The enterprise network for Harbor Lane Legal Group was successfully designed and implemented using Cisco Packet Tracer.

The project achieved all required objectives:

- Department-wise VLAN segmentation was implemented.
- Efficient VLSM addressing was applied.
- Router-on-a-Stick enabled inter-VLAN communication.
- WAN connectivity between offices was established.
- Voice VLAN and DHCP services were configured successfully.
- Network connectivity was verified through testing.

The implemented solution provides a secure, scalable, and manageable network infrastructure suitable for future organizational expansion.

22. Future Improvements

The network can be further enhanced by implementing:

- Firewall security
- Access Control Lists (ACLs)
- Wireless network infrastructure
- Network monitoring tools
- Redundant links
- Dynamic routing protocols such as OSPF

These improvements can increase reliability, security, and performance.

End of Report
